INFORMATION TO KEEP WITH YOU

My complete treatment handbook

Dose-dense AC → paclitaxel every 2 weeks

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
14 days
Intervals differ by treatment phase

Treatment stages
2
Explained in the treatment reference

Regimen source
NCCP 00278
Version 7

> Four AC cycles 14 days apart, followed by four paclitaxel doses 14 days apart. Growth-factor support accompanies the dose-dense course.

Treatment stages

01 **Phase 1 · AC**
14-day interval

02 **Phase 2 · Paclitaxel**
14-day interval

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name _____

Hospital / oncology team _____

> Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

SACTCHECK · CHAPTER 01

Understanding your treatment

Dose-dense AC → paclitaxel every 2 weeks · National Cancer Control Programme (NCCP) 00278

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Report new breathing symptoms, severe diarrhoea, inability to drink, painful swollen limbs, unusual bleeding, or worsening numbness. Tell staff immediately about infusion-site pain or an allergic reaction.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

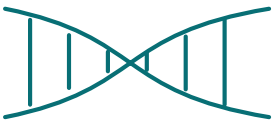
This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

How the medicines work

Part of an individual breast cancer plan to reduce the risk of recurrence, and to shrink the cancer before surgery where that is the agreed setting.

ANTHRACYCLINE CHEMOTHERAPY

Doxorubicin



Interferes with deoxyribonucleic acid (the cell's genetic material) (DNA) handling

Doxorubicin interferes with enzymes and DNA processes needed for cell division. It can also affect heart muscle, so the team considers previous exposure and cardiac risk.

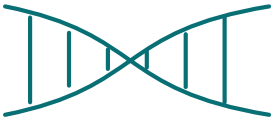
DNA handling disrupted → Cell damage → Division reduced

What to watch for

Report pain or burning at the infusion site immediately. New breathlessness or chest symptoms need prompt review. Red urine for a short time can be a drug-colour effect; pain, clots or persistent bleeding still need advice.

ALKYLATING CHEMOTHERAPY

Cyclophosphamide



Damages DNA needed for growth

Cyclophosphamide is activated in the body and damages DNA. This helps stop dividing cancer cells but can also affect blood-cell production and reproductive tissues.

Drug activated → DNA damage → Cell division affected

What to watch for

Use the prescribed anti-sickness and hydration plan. Report fever, painful urination or blood in the urine.

TAXANE CHEMOTHERAPY

Paclitaxel

Taxol (conventional paclitaxel)



Locks the cell division machinery

Cells use tiny structures called microtubules to separate their chromosomes when dividing. Paclitaxel stabilises these structures so they cannot rearrange normally, interrupting cell division. It can also affect nerves and other healthy tissues.

Microtubules form → Structures remain locked → Division is interrupted

What to watch for

Report tingling or numbness before the next next dose, especially if it affects walking or hand use. Tell the nurse immediately about flushing, chest or throat tightness, itching, dizziness or breathing changes during infusion.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Why this treatment?

Dose-dense AC followed by paclitaxel gives the planned chemotherapy at two-week intervals with growth-factor support. It may suit selected higher-risk early breast cancer. Fitness, blood-count recovery, heart health and practical support matter; it is a schedule choice with additional monitoring.

Your treatment setting

neoadjuvant or adjuvant high-risk breast cancer

The sequence

Four AC cycles 14 days apart, followed by four paclitaxel doses 14 days apart. Growth-factor support accompanies the dose-dense course.

Alternatives and your priorities

Ask what alternatives fit your cancer, what the treatment aims to achieve, and how its practical demands fit your life. The team can discuss personal benefits and risks.

Why this treatment may be chosen

Show trial results and what the numbers mean

neoadjuvant or adjuvant high-risk breast cancer

- Pembrolizumab + chemotherapy
- Placebo + chemotherapy

Your team can discuss expected benefit and alternatives in the context of your own cancer and priorities.

SACTCHECK · CHAPTER 02

Your schedule, tests and appointments

Dose-dense AC → paclitaxel every 2 weeks · NCCP 00278

The phases and how they connect

Treatment stages

01 **Phase 1 · AC**
14-day interval

02 **Phase 2 · Paclitaxel**
14-day interval

Your team confirms the active stage and dates. The full treatment guide explains each step.

This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

14-DAY INTERVAL
Phase 1 · AC

4 cycles

Four AC cycles 14 days apart. Growth-factor support is part of this dose-dense schedule.

14-DAY INTERVAL
Phase 2 · Paclitaxel

4 fortnightly taxane doses

Four AC cycles 14 days apart, followed by four paclitaxel doses 14 days apart. Growth-factor support accompanies the dose-dense course.

The treatment phase determines the medicines, supportive prescription and tests. A phase change does not reset previous drug exposure or cardiac monitoring.

BASELINE → TREATMENT → FOLLOW-UP

When checks happen

Relative milestones, not equal time intervals. The clinical team owns booking, result review and action on overdue tests. This is an educational map, not an automated appointment or result tracker.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00278 v7 · official regimen · ESMO 2024 early breast cancer guideline · treatment and follow-up · ESC 2022 cardio-oncology guideline · cardiac surveillance and risk · NICE NG101 · bone health and follow-up · ESMO metastatic breast cancer guideline · 2026

THE WHOLE TREATMENT JOURNEY

Dose-dense AC → paclitaxel every 2 weeks: treatment and checks

neoadjuvant or adjuvant high-risk breast cancer

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	Early cycles	Later cycles / transition	Next treatment / completion	Follow-up
Cancer medicines NCCP 00278	Agree aim, indication and medicines	Dose-dense AC → paclitaxel every 2 weeks starts	Change from AC to paclitaxel; select new support script	Complete planned course / next decision	Ongoing review / subsequent plan
Surgery EARLY	Adjuvant: surgery already completed	Neoadjuvant: chemotherapy first	Review response / surgical plan	Surgery after neoadjuvant therapy, if applicable	Pathology and recovery guide next steps
Radiotherapy EARLY	Assess indication	Usually after cytotoxic chemotherapy	Arrange sequential timing	Radiation team confirms start	Review effects / follow-up
Blood tests NCCP 00278	full blood count (FBC), kidney and liver profile	Review before chemotherapy cycles	Check again at any phase change	Follow active medicines and abnormalities	Follow unresolved abnormalities
Echo / heart NCCP 00278 · ESC	Baseline echo (ESC); assess risk	High cardiac risk: repeat after 2 anthracycline cycles	If continuing: repeat after 4 cycles at high risk	Review cardiac symptoms and total lifetime exposure	Echo within 12 months; high risk within 3 months
electrocardiogram (ECG) NCCP 00278	Baseline ECG in the NCCP anthracycline source; repeat for symptoms or clinical risk. An ECG does not replace an echo.	Symptoms / risk-based repeat	Review if cardiac concern	ECG and echo answer different questions	Clinical indication
Cancer imaging NCCP 00278	Record baseline extent as appropriate	Clinical response / symptoms review	If before surgery: plan response review	Arrange selected surgery / next treatment	Planned breast imaging; no routine computed tomography (CT)/positron emission tomography (PET)

Care / check	Before starting	Early cycles	Later cycles / transition	Next treatment / completion	Follow-up
Markers / bone health NCCP 00278	Tests selected for active medicines	No routine tumour-marker clearance	No routine blood tumour markers when well	dual-energy X-ray absorptiometry (bone density) (DEXA) if endocrine/ bone-risk indication	Follow the endocrine and bone-health plan
Endocrine / bone BONE · EARLY	Confirm hormone receptors	Plan endocrine therapy when appropriate	Separate from this chemotherapy prescription	Start/continue per multidisciplinary team (MDT); DEXA when indicated	Risk-based bone review

What each check is for

1 **Before first dose**

Baseline

Baseline FBC, kidney and liver tests; review before each AC cycle and each paclitaxel administration under the regimen and local testing plan.

Ask / act: Confirm tests and consent before prescribing.

2 **Heart-monitoring plan**

Baseline → cycles → after treatment

ESC supplementary guidance: baseline echo before anthracycline treatment. High/very-high cardiac risk: echo every two anthracycline cycles (for example before cycles 3 and 5 when continuing), and within 3 months after completion. All adults: echo within 12 months after completion. Other repeat tests depend on risk, cumulative exposure and symptoms.

Ask / act: Name the team responsible for booking and reviewing results.

3

Treatment-phase review

Before each transition

Four AC cycles 14 days apart, followed by four paclitaxel doses 14 days apart. Growth-factor support accompanies the dose-dense course.

Ask / act: Confirm active medicines, support script and the next test checkpoint.

4

Response and follow-up

Planned review points

For early breast cancer, breast imaging and clinical review follow the agreed pathway. Routine CT/PET scans and blood tumour markers are not used to look for distant recurrence when there are no symptoms or clinical concerns.

Ask / **act:** Review symptoms and imaging together; do not change treatment on tumour markers alone.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Your practical treatment plan

STEP 1

Before the first visit

Blood tests, relevant cardiac assessment, medicines review, fertility discussion and the consent conversation come first.

STEP 2

Plan venous access

A cannula, peripherally inserted central catheter (PICC) or implanted port may be used. If a port is chosen, a trained team inserts it in a separate procedure with local anaesthetic and sometimes sedation; the team explains fasting, blood-thinner and wound instructions.

STEP 3

Infusion day

Staff check the line, give prescribed premedicines and treatment, and observe for reactions. Tell them immediately about burning, pain, rash or breathlessness.

STEP 4

Home and next review

This regimen does not routinely need a take-home infusion pump. Staff disconnect the infusion; a port is de-accessed when appropriate. Keep line-care and next-visit instructions.

Port insertion and removal

The line team explains placement, dressing care and when normal activities can resume. A port is accessed with a special needle during treatment. Removal of the implanted device is a separate planned procedure when no longer needed; do not remove or adjust it yourself.

PICC care and urgent line symptoms

A PICC usually needs regular dressing and flushing appointments even between treatments. Fever, redness, swelling, pain, leakage or damage requires prompt advice. Your hospital supplies the exact care instructions.

Visit time and transport

First visits can take longer because of loading doses, premedicines and observation. Ask the unit about expected time, transport and any escort needed after sedation or line insertion.

Phase-specific visit pattern

Infusion and observation times differ by drug and first versus later doses; the unit confirms the full visit duration, including blood review and preparation.

Checks across treatment

Your team arranges, reviews and acts on tests; an overdue result needs follow-up.

Blood tests

Baseline FBC, kidney and liver tests; review before each AC cycle and each paclitaxel administration under the regimen and local testing plan.

Heart monitoring

ESC supplementary guidance: baseline echo before anthracycline treatment. High/very-high cardiac risk: echo every two anthracycline cycles (for example before cycles 3 and 5 when continuing), and within 3 months after completion. All adults: echo within 12 months after completion. Other repeat tests depend on risk, cumulative exposure and symptoms.

ECG

Baseline ECG in the NCCP anthracycline source; repeat for symptoms or clinical risk. An ECG does not replace an echo.

Cancer scans

For early breast cancer, breast imaging and clinical review follow the agreed pathway. Routine CT/PET scans and blood tumour markers are not used to look for distant recurrence when there are no symptoms or clinical concerns.

Surgery, radiotherapy and later treatment

Adjuvant pathway: surgery → chemotherapy → radiotherapy if indicated. Neoadjuvant pathway, where this regimen is indicated: chemotherapy → surgery → selected postoperative treatment and radiotherapy. Lumpectomy versus mastectomy is a surgical/MDT decision.

Hormone-receptor-positive disease

Endocrine treatment is considered when appropriate for the cancer biology and setting. It has its own schedule and monitoring; it is not automatically part of this chemotherapy prescription.

Bone health

DEXA is linked to an aromatase inhibitor, ovarian suppression, treatment-induced menopause or other bone risk—not automatically to every chemotherapy course. The endocrine team confirms baseline testing and risk-based repeat scans.

Markers and follow-up

After local treatment, the team plans mammography of remaining breast tissue. A mastectomy side does not need routine mammography. Report new symptoms rather than waiting for a scan.

Genomic tests

Where appropriate, the MDT may discuss a validated genomic test to inform the chemotherapy decision. This regimen card does not calculate an Oncotype result or select treatment from a score.

Symptoms and supportive care

Dose-dense AC → paclitaxel every 2 weeks · NCCP 00278

AC · 14-day reference cycle

Day 1 → Day 14: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

AC · 14-day reference cycle

Treatment or check	Timing in this view
Treatment · Day 1	 **Day 1** intravenous (IV)
Administration reaction	 **Days 1–14** Report during or after any visit
Nausea / appetite	 **Days 1–5** May occur early or later
Blood counts / infection	 **Days 1–14** Risk varies; fever needs prompt advice
Fatigue / mouth changes	 **Days 1–14** Variable timing and severity

No dates to enter: The chart explains the pattern. Your team books actual appointments and reviews results.

Treatment delays: A delayed dose does not automatically delay an echo or other test that is already due.

Phase changes: AC and taxane prescriptions differ.

Why these bands stay broad: These broad symptom bands are educational possibilities, not measured risk curves. Infusion days are the planned schedule; delays require a new clinical plan. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00278 v7 · official regimen ↗

There are no guaranteed "safe days". Fever, shivers, sudden illness, severe breathing changes or chest pain need prompt assessment on any day.

Phase 2 · Paclitaxel

Day 1 → Day 14: a general pattern

Phase 2 · Paclitaxel

Treatment or check	Timing in this view
Chemotherapy · Day 1	 **Day 1** IV
Administration reaction	 **Days 1–14** Report during or after any visit
Nausea / appetite	 **Days 1–5** May occur early or later
Blood counts / infection	 **Days 1–14** Risk varies; fever needs prompt advice
Fatigue / mouth changes	 **Days 1–14** Variable timing and severity
Nerves / skin / nails	 **Days 1–14** Can build across doses

No dates to enter: The chart explains the pattern. Your team books actual appointments and reviews results.

Treatment delays: A delayed dose does not automatically delay an echo or other test that is already due.

Phase changes: AC and taxane prescriptions differ.

Why these bands stay broad: These broad symptom bands are educational possibilities, not measured risk curves. Infusion days are the planned schedule; delays require a new clinical plan. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00278 v7 · official regimen ↗

Between-treatment notes

Day 1–14: Your current reference cycle

Four AC cycles 14 days apart, followed by four paclitaxel doses 14 days apart. Growth-factor support accompanies the dose-dense course. Follow the team's appointment and medicine instructions; contact them about missed doses or new symptoms.

Support around the medicines

1  **The correct phase prescription**

AC is high emetogenic risk; a separate low-risk taxane script replaces it.

2 **Premedicines and growth factors**

Paclitaxel premedicines differ between weekly and fortnightly dosing. Use the phase-specific prescription.

3 **Tests and appointments**

The team should name the booking owner, due window and person reviewing each required test. Tell them if an expected appointment has not arrived.

4 **After chemotherapy**

Discuss recovery, fertility, rehabilitation and the next treatment or surveillance step.

Questions worth asking

When are heart checks due?

Baseline cardiac assessment before AC. Repeat imaging according to symptoms, anthracycline exposure and cardiac risk; no universal 12-week echo schedule is assigned to AC alone. ESC guidance supplements the source: baseline echo before anthracyclines; high / very-high cardiac risk may need echo every two cycles and within 3 months after treatment. Test timing follows the treatment, clinical findings and local pathway.

Where do surgery and radiotherapy fit?

Two possible pathways: neoadjuvant systemic treatment → surgery / pathology → postoperative plan; or surgery first → adjuvant systemic treatment → radiotherapy when indicated. HER2-only treatment, if prescribed, may overlap radiation / endocrine therapy under MDT review. These are alternatives, not two consecutive surgery courses.

Do I need routine CT scans or tumour markers?

Early breast follow-up uses planned breast imaging. Routine CT / PET, blood tumour markers or other distant-recurrence screening are not recommended without symptoms or a clinical indication. After local treatment: annual mammography of remaining breast tissue; additional imaging when indicated.

Does a port mean a pump at home?

No. This regimen has no routine take-home infusion pump. A port is an access device; the team removes the needle after the visit when appropriate and arranges device removal separately.

Who arranges the scans and tests?

Your treating service should book the planned checks, review the results and explain any change. If an expected appointment is missing, contact the team; this map does not book tests or confirm that they have happened.

> **Try explaining the plan back:** Can you explain which medicines you are receiving now? What symptoms mean you should call urgently? Which blood tests, scans or heart checks come next, and who arranges them?

Doxorubicin

- Low counts, infection and mouth ulcers
- Nausea, hair loss and fatigue
- Heart damage, sometimes delayed
- Severe tissue injury if the infusion leaks
- Rare later blood cancers

Cyclophosphamide

- Low counts and infection
- Nausea, vomiting and hair loss
- Bladder irritation / bleeding
- Fertility effects and menopause

Paclitaxel

- Low blood counts and infection
- Numbness / tingling, sometimes lasting
- Hair loss, muscle or joint pain
- Infusion reaction
- Mouth, bowel, skin and nail effects

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

AC · 14-day schedule

Active medicines: Doxorubicin + Cyclophosphamide

Emetogenic risk: High · AC

Medicine	Dose / route	When
Aprepitant	125 mg orally	Day 1
Ondansetron	16 mg orally	Day 1 Individual dose / QT / hepatic review
Dexamethasone	12 mg orally	Day 1 No routine Days 2–4 dexamethasone in this NCCP AC antiemetic row
Olanzapine	5 mg orally	Days 1–4 Review sedation, falls, interactions and glucose
Aprepitant	80 mg orally daily	Days 2–3
Metoclopramide	10 mg orally up to three times daily	Subsequent days: duration and use to be confirmed Caution with olanzapine: additive extrapyramidal effects; adjust / avoid for individual contraindications

Additional phase-specific support

- granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) is part of each dose-dense AC cycle; prescribe product and timing separately.
- This AC antiemetic script is replaced at the taxane transition.
- At the AC-to-paclitaxel transition, stop using this AC antiemetic template and select the taxane phase. Review residual symptoms separately.
- Dose-dense treatment requires the prescribed G-CSF plan. NCCP requires G-CSF with all cycles and refers to local policy; specify the chosen product, dose and days in the individual prescription.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

Taxane / combination phase

Active medicines: Paclitaxel

Emetogenic risk: Low

Medicine	Dose / route	When
Ondansetron	8 mg orally	On treatment day Review QT, constipation and hepatic dosing; reconcile any steroid already prescribed

Medicine	Dose / route	When
Metoclopramide	10 mg orally up to three times daily as needed	Subsequent days: duration to be confirmed Assess contraindications, renal function and interactions

Additional phase-specific support

- Fortnightly paclitaxel: dexamethasone 20 mg orally at 12 and 6 hours before treatment, OR the source intravenous (IV) option 30 minutes before; chlorphenamine 10 mg IV and famotidine 20 mg IV before treatment. The team selects the route / timing; do not substitute the weekly 8 mg first-dose schedule.
- Growth-factor support is required for the dose-dense taxane phase. Consultant-directed steroid reduction / omission must account for reactions and interactions; if aprepitant is added, the source discusses 12 mg treatment-day dexamethasone.
- Dose-dense treatment requires the prescribed granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) plan. NCCP requires G-CSF with all cycles and refers to local policy; specify the chosen product, dose and days in the individual prescription.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

NCCP 00278 · VERSION 7

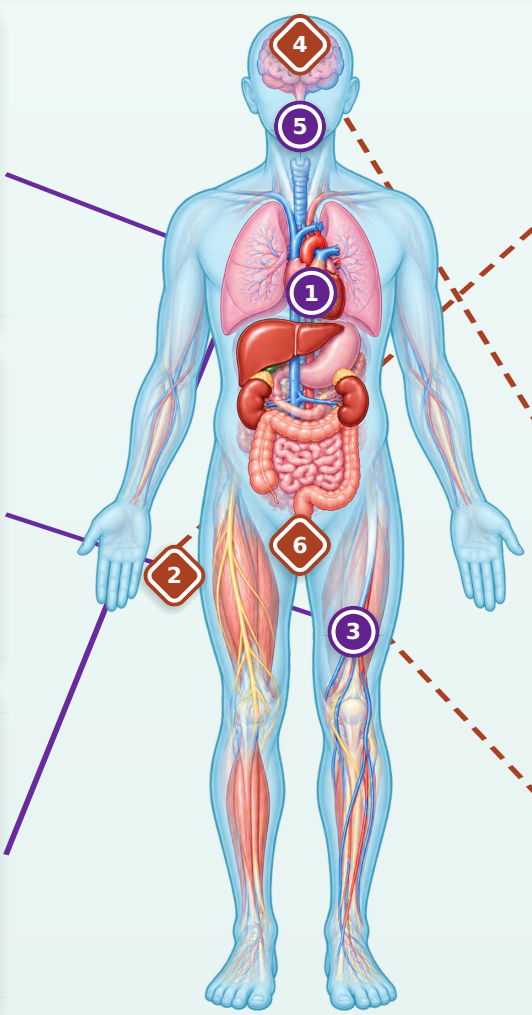
Dose-dense AC → paclitaxel every 2 weeks

Four AC cycles 14 days apart, followed by four paclitaxel doses 14 days apart. Growth-factor support accompanies the dose-dense course.

Source reconciled; independent clinical, pharmacy and patient/PPI review pending. Selected effects; not a complete medicine leaflet.

A picture of side effects

Numbered symptom boxes connect to the anatomical illustration. The medicine responsible may be uncertain; report the symptom.



1 Heart

Doxorubicin

Baseline assessment; report chest pain, breathlessness or swelling.

2 Hands / feet

Paclitaxel

Report numbness or tingling before the next dose.

3 Blood / infection

Chemotherapy phase

Fever, shivers or suddenly unwell: call immediately.

4 Hair / scalp

Chemotherapy phase

Hair loss is common. Ask about scalp cooling before treatment where suitable.

5 Mouth

Chemotherapy phase

Painful ulcers or difficulty drinking need advice.

6 Bladder

Cyclophosphamide

Report blood in urine or painful urination.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Selected medicine-specific effects are highlighted. The map is not a complete list; persistent or new symptoms need review even after a medicine stops.

> Fever, shivers or suddenly unwell: contact oncology immediately. Severe chest pain, breathing difficulty, collapse or seizure: 112 / 999.

SACTCHECK · CHAPTER 04

Sources and further information

Dose-dense AC → paclitaxel every 2 weeks · National Cancer Control Programme (NCCP) 00278

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **Regimen selection and source references**
   https://healthservice.hse.ie/documents/6696/278_v7_Dose_DenseAC-T_Pacli_14_day.pdf
2. **ESMO 2024 early breast cancer guideline · treatment and follow-up**
   https://www.annalsofoncology.org/article/S0923-7534(23)05104-9/fulltext
3. **ESC 2022 cardio-oncology guideline · cardiac surveillance and risk**
   https://academic.oup.com/eurheartj/article/43/41/4229/6673995
4. **NICE NG101 · bone health and follow-up**
   https://www.nice.org.uk/guidance/ng101/chapter/Recommendations
5. **ESMO metastatic breast cancer guideline · 2026**
   https://www.annalsofoncology.org/article/S0923-7534(26)00902-6/fulltext
6. **eviQ · implanted port patient information**
   https://www.eviq.org.au/p/507
7. **eviQ · peripherally inserted central catheter (PICC) patient information**
   https://www.eviq.org.au/p/543
8. **HSE · chemotherapy side effects and urgent advice**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
9. **NOGG · section 7, aromatase inhibitors and fracture risk**
   https://www.nogg.org.uk/full-guideline/section-7-strategies-management-osteoporosis-and-fracture-risk
10. **NCCP NCIS antiemetics v8 (2025), Table 1**
    https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf
11. **ASCO · biomarkers in metastatic breast cancer**
    https://ascopubs.org/doi/10.1200/JCO.2015.61.1459